Face Mask Images Clustering using K-Means and Agglomerative Hierarchical Clustering

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Practical Machine Learning – Project 2

1 Introduction

The goal of this project is to explore face mask images from an unsupervised learning perspective. Instead of using labels during training, the objective is to group images based only on their visual features and then analyze how well the resulting clusters correspond to the *WithMask* and *WithoutMask* classes. The dataset used is the Face Mask Detection 12K Images Dataset, publicly available on Kaggle [1].

Two clustering methods were tested: K-Means and Hierarchical Clustering. Both methods were applied using two different feature representations: Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT) combined with Bag of Visual Words (BoVW). The experiments compare the clustering results using both of the representations and also investigate what type of images are grouped together in the resulting clusters.

2 Dataset and Preprocessing

2.1 Dataset

The dataset used in this project is the Face Mask 12K Image Dataset. It contains images divided into two classes: *WithMask* and *WithoutMask*. The dataset is already split into training, validation and sets. The number of images in each set is shown in Table 1.

Table 1: Number of images in each dataset split

Split	**WithMask**	**WithoutMask**	**Total**
Train	5000	5000	10000
Validation	400	400	800
Test	483	509	992

The class labels were not used when fitting the clustering models. They were only used afterward to evaluate and interpret the obtained clusters.

2.2 Preprocessing

The images in the dataset have different sizes, so some preprocessing was necessary before extracting the features.

For the HOG representation, all images were converted to grayscale and resized to 64×64 pixels. Using the same image size ensured that all HOG feature vectors have the same length.

For SIFT, the images were also converted to grayscale, but they were kept at their original size. SIFT detects local keypoints in each image, so resizing was not required for this representation.

3 Feature Representations

3.1 HOG

Histogram of Oriented Gradients (HOG) describes an image using the directions of intensity changes, or gradients. This allows HOG to capture information about edges and shapes present in the image. For this representation, each image was converted to grayscale and resized to 64×64 pixels. HOG features were then extracted using 9 orientations, cells of 8×8 pixels and blocks of 2×2 cells. L2-Hys normalization was used for the blocks. With these settings, each image was represented by a feature vector containing 1764 values.

3.2 SIFT and Bag of Visual Words

SIFT detects distinctive local points in an image and creates a descriptor for each keypoint. Unlike HOG, the number of SIFT descriptors can be different for each image. Thus, the SIFT descriptors cannot be used directly as fixed-sized input for the models.

That is why SIFT was combined with the Bag of Visual Words method. First, the SIFT descriptors were extracted from the training images. Then, a random sample of 100,000 descriptors was used to build a visual vocabulary with K-Means. The vocabulary contained 100 visual words.

For each image, its SIFT descriptors were assigned to the closest visual words. A histogram was created by counting how many descriptors belonged to each visual word. The histogram was normalized, resulting in a feature vector with 100 values for every image.

4 Clustering Methods

Two clustering methods were used in this project: K-Means and Hierarchical Clustering. The models were fitted using only the extracted image features, without using the class labels.

4.1 K-Means

K-Means groups the samples into a predefined number of clusters based on their distance from the cluster centroids. During training, the centroids are updated until the assignments become stable.

Several values for the number of clusters, initialization method and number of initializations were tested. The model was fitted on the training set, while the validation set was used to select the best hyperparameters.

4.2 Hierarchical Clustering

Agglomerative Hierarchical Clustering starts with each sample in its own cluster and gradually merges the closest clusters until the desired number of clusters is reached.

Different numbers of clusters, linkage methods and distance metrics were tested. Ward linkage was used only with Euclidean distance, since this is the distance supported by this linkage method.

Unlike K-Means, Agglomerative Clustering does not provide a `predict()` method for unseen samples. Therefore, clustering was performed separately on the validation and test sets, while keeping the hyperparameters selected on the validation set fixed for the final test evaluation.

4.3 Evaluation Metric

The clustering results were evaluated using the Adjusted Rand Index (ARI). ARI measures the agreement between the clusters produced by the model and the real class labels, while

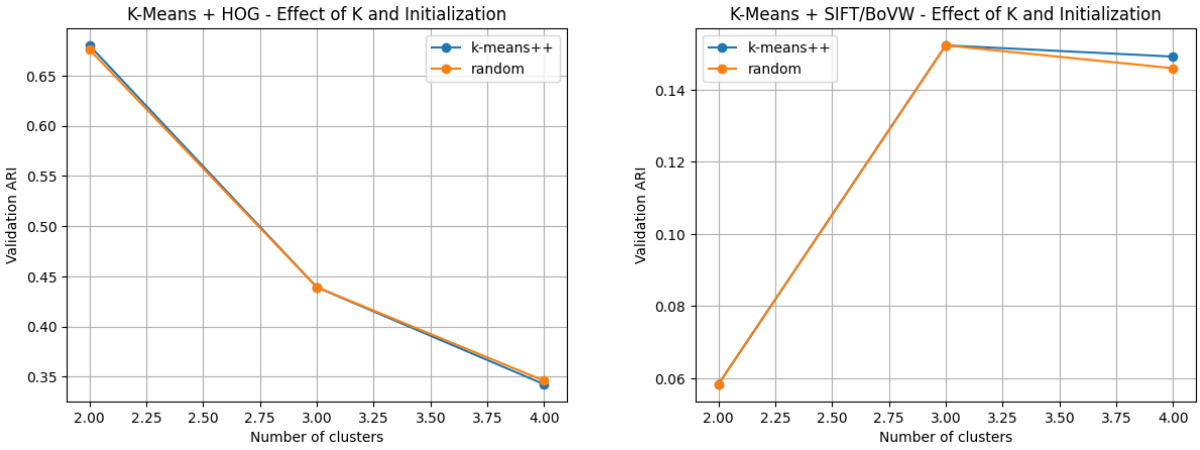
correcting for agreement that could occur by chance. The class labels were used only for this evaluation and were not provided to the clustering models.

5 Results

5.1 Hyperparameter Tuning

5.1.1 K-Means

For K-means, the tuned hyperparameters were: the number of clusters K, the initialization method and the number of initializations (`n_init`). The best configuration was selected based on the validation ARI. Since changing `n_init` had very little influence on the results, the plots focus on the effect of K and the initialization method.



(a) HOG features

(b) SIFT/BoVW features

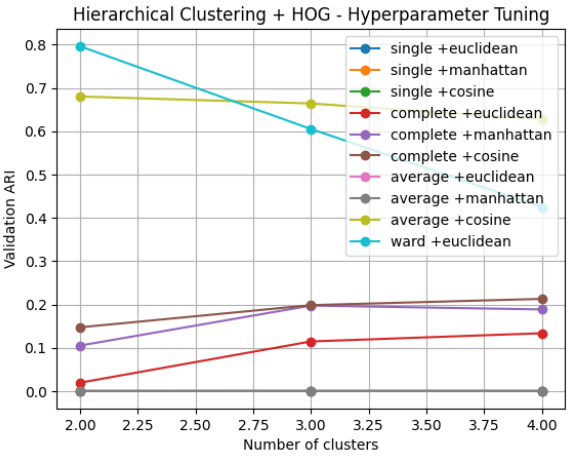
Figure 1: Validation ARI obtained with K-Means for different numbers of clusters and initialization methods.

For HOG features, the best configuration used $K = 2$, `k-means++` initialization and `n_init`=5, obtaining a validation ARI of 0.6802.

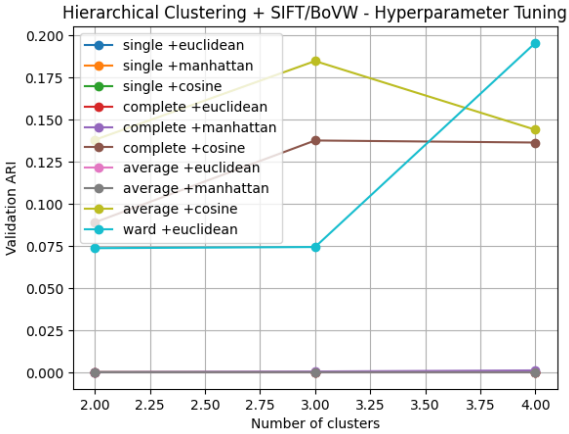
For SIFT/BoVW features, the best configuration used $K = 3$, random initialization and `n_init`=5, obtaining a validation ARI of 0.1541.

5.1.2 Hierarchical Clustering

For Hierarchical Clustering, the tuned hyperparameters were the number of clusters K, the linkage method and the distance metric. The tested linkage methods were single, complete, average and ward, while the tested distance metrics were Euclidean, Manhattan and cosine. Ward linkage was only combined with Euclidean distance.



(a) HOG features



(b) SIFT/BoVW features

Figure 2: Validation ARI obtained with Hierarchical Clustering for different numbers of clusters, linkage methods and distance metrics.

For HOG features, the best configuration used $K = 2$, ward linkage and Euclidean distance, obtaining a validation ARI of 0.7963.

For SIFT/BoVW features, the best configuration used $K = 4$, ward linkage and Euclidean distance, obtaining a validation ARI of 0.1955.

5.2 Final Test Results

After selecting the best hyperparameters using the validation set, the final configurations were evaluated on the test set.

Table 2: Validation and test ARI for the best clustering configurations.

Method	**Features**	**Validation ARI**	**Test ARI**
K-Means	HOG	0.6802	0.6863
K-Means	SIFT/BoVW	0.1541	0.1730
Hierarchical	HOG	0.7963	0.7867
Hierarchical	SIFT/BoVW	0.1955	0.1725

The test results were generally close to the validation results. HOG clearly produced better clustering results than SIFT/BoVW for both methods. The best overall result was obtained by Hierarchical Clustering with HOG features.

5.3 Comparison with Baselines

The clustering results were compared with two baselines. A random clustering baseline was evaluated using ARI, while Logistic Regression was used as a supervised reference model and evaluated using classification accuracy.

The random baseline obtained an ARI close to zero, showing that all tested clustering configurations performed better than random chance. The supervised Logistic Regression model achieved a test accuracy of 0.98085 with HOG features and 0.83367 with SIFT/BoVW features. As expected, the supervised model obtained better results, since it was trained using the class labels.

Table 3: Clustering results compared with random chance.

Method	**Features**	**Test ARI**
Random clustering	–	≈ 0
K-Means	HOG	0.6863
K-Means	SIFT/BoVW	0.1730
Hierarchical	HOG	0.7867
Hierarchical	SIFT/BoVW	0.1725

Table 4: Supervised Logistic Regression baseline.

Features	**Test Accuracy**
HOG	0.98085
SIFT/BoVW	0.83367

An interesting difference can be observed between the supervised and unsupervised settings. In the supervised experiments, SIFT/BoVW performed worse than HOG, but still presented generally good results. In the unsupervised setting, however, the ARI values obtained with SIFT/BoVW are significantly lower.

This suggests that the SIFT/BoVW representation may capture visual similarities that are not strongly related to the *WithMask*/*WithoutMask* distinction. In other words, the clustering models may group the images according to other visual patterns instead of mainly separating them based on the presence of a mask.

6 Cluster Interpretation

6.1 HOG Features

The clusters obtained using HOG features correspond quite closely to the two real classes. For K-Means, Cluster 0 contains mostly *WithMask* images (460 out of 522), while Cluster 1 contains mostly *WithoutMask* images (447 out of 470). Hierarchical Clustering produces an even clearer separation: Cluster 0 contains 472 *WithMask* and 45 *WithoutMask* images, while Cluster 1 contains 464 *WithoutMask* and only 11 *WithMask* images.

The majority examples support the idea that HOG mainly separates the images according to the presence of a mask. The minority examples also show some possible reasons for incorrect grouping. Some images are highly pixelated or blurred, while others are tightly cropped or contain additional edges around the face, such as a hand close to the face. These characteristics can change the edge and shape information captured by HOG and may cause an image to be grouped with the opposite class.



(a) K-Means + HOG (b) Hierarchical Clustering + HOG

Figure 3: Majority and minority examples from the HOG-based clusters.

6.2 SIFT/BoVW Features

For SIFT/BoVW, the selected configurations produced three clusters with K-Means and four clusters with Hierarchical Clustering. Example images from each cluster were inspected in order to look for common visual characteristics.

Unlike the HOG-based clusters, no clear structure related to the *WithMask*/*WithoutMask* distinction could be observed. The clusters appeared more mixed, and in most cases there was no obvious visual rule that explained why the images were grouped together. One K-Means cluster contained several low-resolution or strongly pixelated images, suggesting that image quality may have influenced the grouping to some extent.

7 Conclusion

This project compared K-Means and Hierarchical Clustering using two different feature representations, HOG and SIFT/BoVW, for grouping face mask images.

The experiments showed that HOG produced clusters that matched the real *WithMask*/*WithoutMask* classes much better than SIFT/BoVW. Hierarchical Clustering with HOG achieved the strongest clustering result overall, while the SIFT/BoVW representations produced more mixed clusters and lower ARI values.

References

[1] A. Jangra, *Face Mask Detection ∼12K Images Dataset*, Kaggle, accessed September 2026. Available online: `https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset`

[2] R. Kumari, P. Pallavi and P. Saurabh, *Enhancing face mask detection performance using stacked neural ensembles in mask fusion*, Discover Artificial Intelligence, vol. 6, article 171, 2026. doi: 10.1007/s44163-025-00826-4.